

Your grade: 100%

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Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1.

True or False:

An improper prior distribution is a probability distribution that has a negative mass or density value.

1 / 1 point

True

False

Nice work

Nice!
2.

In order to use an improper prior distribution, one must:

1 / 1 point

Verify that the improper prior distribution leads to a proper posterior distribution.

Verify that the likelihood integrates to one.

Explicitly compute the denominator of Bayes' theorem.

None of the above

Nice work

Nice!
3.

True or False:

The uniform distribution (discrete or continuous) is an attempt to codify "indifference" about a parameter in Bayesian inference.

1 / 1 point

True

False

Nice work

Nice!
4.

True or False:

Consider the beta-binomial conjugate family. The prior $\text{Beta}(3,2)$ is a Jeffreys' prior.

1 / 1 point

True

False

Nice work

Nice!
5.

Select all that apply:

Consider the problem of inferring the mean, μ , of a normal distribution with known variance. Which of the following are considered Jeffreys' priors?

1 / 1 point

$\pi(\mu) = 10$

Nice work

Nice!

Any normal distribution with finite mean and variance.

$\pi(\mu) = 1$

Nice work

Nice!

$N(0, 1)$

None of the above.

6.

True or False:

The Poisson distribution has no Jeffreys' prior.

1 / 1 point

True

False

Nice work

Nice!

7.

True or False:

Jeffreys' prior is designed to be invariant under reparameterization of the model.

1 / 1 point

True

False

Nice work

Nice!

8.

What is the main **disadvantage** of using improper priors in Bayesian inference?

1 / 1 point

They do not depend on the likelihood function.

They always lead to incorrect inference.

They can sometimes lead to improper posteriors.

They cannot be used in any Bayesian analysis.

Nice work

Nice!

9.

Which of the following is an example of an improper prior?

1 / 1 point

$\text{Beta}(1,1)$ prior for a Bernoulli parameter

$N(0,1)$ prior for a normal mean

$P(\lambda) \propto 1/\lambda$ for a Poisson rate λ

$\text{Gamma}(2,2)$ prior for an exponential rate

Nice work

Correct! This function does not integrate to 1.

10.

Which of the following priors would typically be used as an **objective prior** for a Bernoulli parameter θ ?

1 / 1 point

$N(0,1)$

$\text{Beta}(5,5)$

$\text{Beta}(2,2)$

$\text{Beta}(1/2, 1/2)$

Nice work

Nice!

11.

True or False:

The choice of an objective prior is entirely independent of the likelihood function.

1 / 1 point

True

False

Nice work

Nice! An objective prior is defined in terms of the Jeffreys prior, which makes use of the likelihood function.